Final Project

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Read in the cogimp0 dataset

```
# Access and read the cogimp0 file from the drive
cogimp0 = drive_download(
  as_id("1xDBY_8-djIWfKWTBKrxlYbZ9mkw6zlJ2"),
  path = tempfile(fileext = ".sav"),
  overwrite = TRUE
) %>%
  { read_sav(.$local_path) }
```

```
## File downloaded:

## * 'COGIMP9220A_R.sav' <id: 1xDBY_8-djIWfKWTBKrxlYbZ9mkw6zlJ2>

## Saved locally as:

## * '/var/folders/8_/p0sgn35n70bcqqr1rc6jxwyr0000gn/T//Rtmpt6gCGZ/filee0c97a82b617.sav'
```

```
# generate codebook dataframe
cogimp0_codebook = labelled::generate_dictionary(cogimp0)
```

Read in the 2022 tracker file dataset

```
# read in the trk2022v2 file
trk2022v2 = drive_download(
  as_id("1GOwZriMuTMftZYP834aJG7nMrN3k4T5E"),
  path = tempfile(fileext = ".sav"),
  overwrite = TRUE
) %>%
  { read_sav(.$local_path) }
```

```
## File downloaded:

## * 'trk2022tr_r.sav' <id: 1GOwZriMuTMftZYP834aJG7nMrN3k4T5E>
```

```
## Saved locally as:

## * '/var/folders/8_/p0sgn35n70bcqqr1rc6jxwyr0000gn/T//Rtmpt6gCGZ/filee0c929a6035c.sav'
```

```r
# generate codebook
trk2022v2_codebook = labelled::generate_dictionary(trk2022v2)
```

Combine the data and variables we need

```r
cogimp0   <- cogimp0 %>% mutate(hhid = as.character(HHID), pn = as.character(PN))
trk2022v2 <- trk2022v2 %>% mutate(hhid = as.character(hhid), pn = as.character(pn))
```

```r
data_w13 <- cogimp0 %>%
  dplyr::select(hhid, pn,R13MO, R13DY, R13YR, R13DW, R13SCIS, R13CACT, R13PRES, R13VP) %>%rename(
    Month= R13MO,
    Day= R13DY,
    Year= R13YR,
    DOW= R13DW,
    Scissors= R13SCIS,
    Cactus= R13CACT,
    President= R13PRES,
    VP = R13VP
  ) %>%
  mutate(wave = "W13")
```

```r
cov_w13 <- trk2022v2 %>%
  dplyr::select(hhid, pn, page, gender, degree, race) %>%
  rename(age = page) %>%
  mutate(wave = "W13")
```

```r
lca_raw <- data_w13 %>%
  inner_join(cov_w13, by = c("hhid", "pn", "wave"))

indicator_names <- c("Month","Day","Year","DOW","Scissors","Cactus","President","VP")
```

```r
lca_ready <- lca_raw %>%
  dplyr::mutate(across(all_of(indicator_names),
                       ~ ifelse(is.na(.), NA, as.integer(. + 1))))
```

```r
summary(lca_raw[ , indicator_names])
```

```
##      Month             Day              Year             DOW
##  Min.   :0.000   Min.   :0.000   Min.   :0.000   Min.   :0.000
##  1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
##  Median :1.000   Median :1.000   Median :1.000   Median :1.000
##  Mean   :0.959   Mean   :0.808   Mean   :0.958   Mean   :0.958
##  3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:1.000
##  Max.   :1.000   Max.   :1.000   Max.   :1.000   Max.   :1.000
```

```
##  NA's   :27341   NA's   :27341   NA's   :27341   NA's   :27341
##     Scissors         Cactus        President           VP
##  Min.   :0.000   Min.   :0.00    Min.   :0.000   Min.   :0.00
##  1st Qu.:1.000   1st Qu.:1.00    1st Qu.:1.000   1st Qu.:0.00
##  Median :1.000   Median :1.00    Median :1.000   Median :1.00
##  Mean   :0.985   Mean   :0.92    Mean   :0.972   Mean   :0.63
##  3rd Qu.:1.000   3rd Qu.:1.00    3rd Qu.:1.000   3rd Qu.:1.00
##  Max.   :1.000   Max.   :1.00    Max.   :1.000   Max.   :1.00
##  NA's   :27341   NA's   :27341   NA's   :27341   NA's   :27341
```

```r
summary(lca_raw$age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    21.0    64.0    92.0   526.1   999.0   999.0
```

```r
table(lca_raw$gender, useNA = "ifany")
```

```
##
##     1     2
## 17830 23211
```

```r
table(lca_raw$degree, useNA = "ifany")
```

```
##
##     0     1     2     3     4     5     6     9
## 10117  2015 17600  2041  4771  2418   797  1282
```

```r
table(lca_raw$race,   useNA = "ifany")
```

```
##
##     0     1     2     7
##   111 29627  7871  3432
```

```r
table(lca_raw$wave)
```

```
##
##   W13
## 41041
```

Fit LCA model

Without covariates

```r
set.seed(687)
f_ind <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~ 1
sub_df <- lca_ready %>% dplyr::sample_frac(0.5)
fit_list <- lapply(2:5, function(k) {
  poLCA(f_ind, data = sub_df, nclass = k, maxiter = 1500, nrep = 3, verbose = FALSE)
```

```r
})

data.frame(
  k   = 2:5,
  AIC = sapply(fit_list, \(m) m$aic),
  BIC = sapply(fit_list, \(m) m$bic),
  Gsq = sapply(fit_list, \(m) m$Gsq)
)
```

```
##   k      AIC      BIC      Gsq
## 1 2 27524.67 27640.81 693.3718
## 2 3 27199.26 27376.89 349.9661
## 3 4 27115.02 27354.14 247.7247
## 4 5 27081.23 27381.83 195.9287
```

After comparing AIC and BIC and considering simplicity and efficient, we chose 4 classes due to the lowest BIC and a relatively low AIC.

```r
set.seed(687)
lca_k4 <- poLCA(f_ind, data = lca_ready, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)
```

```
## Model 1: llik = -26936.26 ... best llik = -26936.26
## Model 2: llik = -26936.26 ... best llik = -26936.26
## Model 3: llik = -26936.26 ... best llik = -26936.26
## Model 4: llik = -26936.26 ... best llik = -26936.26
## Model 5: llik = -26936.26 ... best llik = -26936.26
## Model 6: llik = -26936.26 ... best llik = -26936.26
## Model 7: llik = -26936.26 ... best llik = -26936.26
## Model 8: llik = -26936.26 ... best llik = -26936.26
## Model 9: llik = -26936.26 ... best llik = -26936.26
## Model 10: llik = -26936.26 ... best llik = -26936.26
## Model 11: llik = -26936.26 ... best llik = -26936.26
## Model 12: llik = -26936.26 ... best llik = -26936.26
## Model 13: llik = -26936.26 ... best llik = -26936.26
## Model 14: llik = -26936.26 ... best llik = -26936.26
## Model 15: llik = -26936.26 ... best llik = -26936.26
## Model 16: llik = -26936.26 ... best llik = -26936.26
## Model 17: llik = -26936.26 ... best llik = -26936.26
## Model 18: llik = -26936.26 ... best llik = -26936.26
## Model 19: llik = -26936.26 ... best llik = -26936.26
## Model 20: llik = -26936.26 ... best llik = -26936.26
## Conditional item response (column) probabilities,
##  by outcome variable, for each class (row)
##
## $Month
##            Pr(1)  Pr(2)
## class 1:  0.0102 0.9898
## class 2:  0.7043 0.2957
## class 3:  0.3854 0.6146
## class 4:  0.0356 0.9644
##
## $Day
```

```
##            Pr(1)  Pr(2)
## class 1:  0.1261 0.8739
## class 2:  0.9322 0.0678
## class 3:  0.7832 0.2168
## class 4:  0.2831 0.7169
##
## $Year
##            Pr(1)  Pr(2)
## class 1:  0.0069 0.9931
## class 2:  0.8625 0.1375
## class 3:  0.3461 0.6539
## class 4:  0.0463 0.9537
##
## $DOW
##            Pr(1)  Pr(2)
## class 1:  0.0113 0.9887
## class 2:  0.6371 0.3629
## class 3:  0.3049 0.6951
## class 4:  0.0624 0.9376
##
## $Scissors
##            Pr(1)  Pr(2)
## class 1:  0.0058 0.9942
## class 2:  0.1112 0.8888
## class 3:  0.0132 0.9868
## class 4:  0.0476 0.9524
##
## $Cactus
##            Pr(1)  Pr(2)
## class 1:  0.0253 0.9747
## class 2:  0.4581 0.5419
## class 3:  0.0613 0.9387
## class 4:  0.2969 0.7031
##
## $President
##            Pr(1)  Pr(2)
## class 1:  0.0000 1.0000
## class 2:  0.6065 0.3935
## class 3:  0.0315 0.9685
## class 4:  0.0953 0.9047
##
## $VP
##            Pr(1)  Pr(2)
## class 1:  0.2406 0.7594
## class 2:  1.0000 0.0000
## class 3:  0.5197 0.4803
## class 4:  0.8702 0.1298
##
## Estimated class population shares
##  0.7772 0.0183 0.0379 0.1665
##
## Predicted class memberships (by modal posterior prob.)
##  0.8693 0.0176 0.0315 0.0817
##
```

```
## ==========================================================
## Fit for 4 latent classes:
## ==========================================================
## number of observations: 13700
## number of estimated parameters: 35
## residual degrees of freedom: 220
## maximum log-likelihood: -26936.26
##
## AIC(4): 53942.53
## BIC(4): 54205.91
## G^2(4): 312.577 (Likelihood ratio/deviance statistic)
## X^2(4): 350.7069 (Chi-square goodness of fit)
##
```

```
lca_k4$probs
```

```
## $Month
##                Pr(1)     Pr(2)
## class 1:  0.01021639 0.9897836
## class 2:  0.70425104 0.2957490
## class 3:  0.38540730 0.6145927
## class 4:  0.03557407 0.9644259
##
## $Day
##               Pr(1)      Pr(2)
## class 1:  0.1261458 0.87385418
## class 2:  0.9322472 0.06775285
## class 3:  0.7831913 0.21680871
## class 4:  0.2830916 0.71690836
##
## $Year
##                Pr(1)     Pr(2)
## class 1:  0.00694969 0.9930503
## class 2:  0.86246217 0.1375378
## class 3:  0.34613287 0.6538671
## class 4:  0.04630827 0.9536917
##
## $DOW
##                Pr(1)     Pr(2)
## class 1:  0.01129819 0.9887018
## class 2:  0.63709223 0.3629078
## class 3:  0.30494914 0.6950509
## class 4:  0.06238342 0.9376166
##
## $Scissors
##                 Pr(1)     Pr(2)
## class 1:  0.005780905 0.9942191
## class 2:  0.111216129 0.8887839
## class 3:  0.013239550 0.9867604
## class 4:  0.047631039 0.9523690
##
## $Cactus
##                Pr(1)     Pr(2)
## class 1:  0.02525382 0.9747462
```

```
## class 2:  0.45806410 0.5419359
## class 3:  0.06125751 0.9387425
## class 4:  0.29691991 0.7030801
##
## $President
##                    Pr(1)     Pr(2)
## class 1:  5.441065e-119 1.0000000
## class 2:   6.064720e-01 0.3935280
## class 3:   3.149983e-02 0.9685002
## class 4:   9.531016e-02 0.9046898
##
## $VP
##                Pr(1)        Pr(2)
## class 1:  0.2405939 7.594061e-01
## class 2:  1.0000000 1.682929e-29
## class 3:  0.5196684 4.803316e-01
## class 4:  0.8702308 1.297692e-01
```

```r
lca_ready$.rowid <- seq_len(nrow(lca_ready))
mf <- model.frame(f_ind, data = lca_ready, na.action = na.omit)
idx <- as.integer(row.names(mf))
tab <- table(lca_k4$predclass, lca_ready$wave[idx])
tab
```

```
##
##       W13
##   1 11909
##   2   241
##   3   431
##   4  1119
```

With covariates

```r
set.seed(687)
f_cov <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~
  age + factor(gender) + factor(degree) + factor(race)

lca_cov <- poLCA(f_cov, data = lca_ready, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)
```

```
## Model 1: llik = -29364.08 ... best llik = -29364.08
## Model 2: llik = -28236.02 ... best llik = -28236.02
## Model 3: llik = -27896.82 ... best llik = -27896.82
## Model 4: llik = -26803.71 ... best llik = -26803.71
## Model 5: llik = -25591.89 ... best llik = -25591.89
## Model 6: llik = -28170.09 ... best llik = -25591.89
## Model 7: llik = -26722.91 ... best llik = -25591.89
## Model 8: llik = -28550.77 ... best llik = -25591.89
## Model 9: llik = -26758.9 ... best llik = -25591.89
## Model 10: llik = -27445.36 ... best llik = -25591.89
## Model 11: llik = -26758.91 ... best llik = -25591.89
## Model 12: llik = -26758.34 ... best llik = -25591.89
## Model 13: llik = -29517.03 ... best llik = -25591.89
## Model 14: llik = -26765.07 ... best llik = -25591.89
```

```
## Model 15: llik = -26624.58 ... best llik = -25591.89
## Model 16: llik = -26750.32 ... best llik = -25591.89
## Model 17: llik = -25800.14 ... best llik = -25591.89
## Model 18: llik = -25883.55 ... best llik = -25591.89
## Model 19: llik = -26747.44 ... best llik = -25591.89
## Model 20: llik = -26780.55 ... best llik = -25591.89
## Conditional item response (column) probabilities,
##  by outcome variable, for each class (row)
##
## $Month
##            Pr(1)  Pr(2)
## class 1:  0.0516 0.9484
## class 2:  0.4969 0.5031
## class 3:  0.0202 0.9798
## class 4:  0.0156 0.9844
##
## $Day
##            Pr(1)  Pr(2)
## class 1:  0.2050 0.7950
## class 2:  0.8540 0.1460
## class 3:  0.2066 0.7934
## class 4:  0.1401 0.8599
##
## $Year
##            Pr(1)  Pr(2)
## class 1:  0.0249 0.9751
## class 2:  0.5955 0.4045
## class 3:  0.0250 0.9750
## class 4:  0.0100 0.9900
##
## $DOW
##            Pr(1)  Pr(2)
## class 1:  0.0403 0.9597
## class 2:  0.4442 0.5558
## class 3:  0.0396 0.9604
## class 4:  0.0154 0.9846
##
## $Scissors
##            Pr(1)  Pr(2)
## class 1:  0.0769 0.9231
## class 2:  0.0669 0.9331
## class 3:  0.0165 0.9835
## class 4:  0.0066 0.9934
##
## $Cactus
##            Pr(1)  Pr(2)
## class 1:  0.7435 0.2565
## class 2:  0.2900 0.7100
## class 3:  0.0828 0.9172
## class 4:  0.0190 0.9810
##
## $President
##            Pr(1)  Pr(2)
## class 1:  0.0380 0.9620
```

```
## class 2:  0.3481 0.6519
## class 3:  0.0388 0.9612
## class 4:  0.0017 0.9983
##
## $VP
##            Pr(1)  Pr(2)
## class 1:  0.7266 0.2734
## class 2:  0.8419 0.1581
## class 3:  0.9676 0.0324
## class 4:  0.1213 0.8787
##
## Estimated class population shares
##  0.0466 0.048 0.2196 0.6858
##
## Predicted class memberships (by modal posterior prob.)
##  0.0374 0.0437 0.2626 0.6562
##
## =========================================================
## Fit for 4 latent classes:
## =========================================================
## 2 / 1
##                  Coefficient  Std. error  t value  Pr(>|t|)
## (Intercept)         -5.97982     0.50742  -11.785     0.000
## age                  0.09677     0.00816   11.864     0.000
## factor(gender)2     -0.62671     0.17268   -3.629     0.000
## factor(degree)1      0.73673     0.38007    1.938     0.054
## factor(degree)2      0.33426     0.18738    1.784     0.076
## factor(degree)3      1.81543     0.78017    2.327     0.021
## factor(degree)4      6.35419     0.15419   41.209     0.000
## factor(degree)5      6.66894     0.20502   32.528     0.000
## factor(degree)6      4.79474     0.34110   14.057     0.000
## factor(degree)9      1.29222     0.63377    2.039     0.043
## factor(race)1       -0.07024     0.31625   -0.222     0.824
## factor(race)2       -2.28273     0.20348  -11.218     0.000
## factor(race)7       -1.67428     0.20749   -8.069     0.000
## =========================================================
## 3 / 1
##                  Coefficient  Std. error  t value  Pr(>|t|)
## (Intercept)          5.77112     0.51925   11.114     0.000
## age                 -0.04084     0.00691   -5.910     0.000
## factor(gender)2     -0.16583     0.15737   -1.054     0.293
## factor(degree)1      0.63872     0.33526    1.905     0.058
## factor(degree)2      0.39261     0.17143    2.290     0.023
## factor(degree)3      1.55119     0.74864    2.072     0.040
## factor(degree)4      5.51451     0.16862   32.703     0.000
## factor(degree)5      5.16784     0.25361   20.377     0.000
## factor(degree)6      3.72312     0.37506    9.927     0.000
## factor(degree)9      0.89183     0.40801    2.186     0.030
## factor(race)1       -0.22585     0.39710   -0.569     0.570
## factor(race)2       -2.93490     0.32104   -9.142     0.000
## factor(race)7       -2.09596     0.31265   -6.704     0.000
## =========================================================
## 4 / 1
##                  Coefficient  Std. error  t value  Pr(>|t|)
```

```
## (Intercept)         1.66683     0.49148    3.391     0.001
## age                -0.01311     0.00636   -2.062     0.041
## factor(gender)2    -0.51423     0.14732   -3.490     0.001
## factor(degree)1     1.69525     0.32455    5.223     0.000
## factor(degree)2     1.99520     0.16528   12.071     0.000
## factor(degree)3     3.70465     0.72991    5.076     0.000
## factor(degree)4     8.56184     0.10453   81.911     0.000
## factor(degree)5     8.80267     0.14729   59.762     0.000
## factor(degree)6     7.17081     0.22053   32.516     0.000
## factor(degree)9     2.94865     0.39213    7.520     0.000
## factor(race)1       1.92126     0.39754    4.833     0.000
## factor(race)2      -0.96454     0.31296   -3.082     0.002
## factor(race)7      -0.92170     0.30853   -2.987     0.003
## =========================================================
## number of observations: 13700
## number of estimated parameters: 71
## residual degrees of freedom: 184
## maximum log-likelihood: -25591.89
##
## AIC(4): 51325.77
## BIC(4): 51860.06
## X^2(4): 800.1784 (Chi-square goodness of fit)
##
## ALERT: estimation algorithm automatically restarted with new initial values
##
```

```
lca_cov$P
```

```
## [1] 0.04662023 0.04795315 0.21962966 0.68579696
```

```
lca_cov$coeff
```

```
##                        [,1]       [,2]        [,3]
## (Intercept)     -5.97982197  5.7711180  1.66683003
## age              0.09677193 -0.0408377 -0.01310743
## factor(gender)2 -0.62671403 -0.1658345 -0.51423287
## factor(degree)1  0.73672953  0.6387191  1.69524506
## factor(degree)2  0.33426466  0.3926119  1.99519942
## factor(degree)3  1.81542845  1.5511922  3.70464512
## factor(degree)4  6.35418669  5.5145142  8.56183519
## factor(degree)5  6.66893811  5.1678428  8.80267155
## factor(degree)6  4.79474375  3.7231190  7.17081014
## factor(degree)9  1.29222246  0.8918251  2.94865355
## factor(race)1   -0.07023951 -0.2258541  1.92125970
## factor(race)2   -2.28273296 -2.9349019 -0.96454204
## factor(race)7   -1.67428016 -2.0959557 -0.92170367
```